

KENDRIYA VIDYALAYA SANGATHAN (PATNA REGION)
PRE- BOARD -1 (2024-25)
SCIENCE (086) SET- A

CLASS- X

Max. Marks: 80

Time Allowed: 3 Hrs.

General Instructions:-

1. All questions are compulsory. However, an internal choice of 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. Section A would have 16 simple/complex MCQs and 04 Assertion- Reasoning type questions carrying 1 mark each.
3. Section B would have 06 short answer (SA) type questions carrying 02 marks each.
4. Section C would have 07 short answer (SA) type questions carrying 3 marks each.
5. Section D would have 03 long answer (LA) type questions carrying 5 marks each.
6. Section E would have 3 source based/ case based/ integrated units of assessment (04 marks each) with sub- parts of 1/2/3 marks.

Section- A

Questions 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer of these questions.

1. Which of the following statements about the given reaction are correct?



- (i) Iron metal is getting oxidized
- (ii) Water is getting reduced
- (iii) Water is acting as reducing agent
- (iv) Water is acting as oxidizing agent
- (a) (i), (ii) and (iii) (b) (iii) and (iv)
- (c) (i), (ii) and (iv) (d) (ii) and (iv)

2. Match the chemical substances given in column (A) with their appropriate application given in column (B).

- | Column (A) | Column (B) |
|--|---|
| (A) Bleaching powder | (i) Preparation of glass |
| (B) Baking soda | (ii) Production of H_2 and Cl_2 |
| (C) Washing soda | (iii) Decolourisation |
| (D) Sodium chloride | (iv) Antacid |
| (a) A --- (ii), B --- (i), C --- (iv), D --- (iii) | |
| (b) A --- (iii), B --- (ii), C --- (iv), D --- (i) | |
| (c) A --- (iii), B --- (iv), C --- (i), D --- (ii) | |
| (d) A --- (ii), B --- (iv), C --- (i), D --- (iii) | |

3. Aluminium is used for making cooking utensils. Which of the following properties of aluminium are responsible for the same?
- (i) Good thermal conductivity

(ii) Good electrical conductivity

(iii) Ductility

(iv) High melting point

(a) (i) and (ii)

(b) (i) and (iii)

(c) (ii) and (iii)

(d) (i) and (iv)

4. Alloys are homogeneous mixture of a metal with a metal or non-metal. Which among the following alloys contain non-metal as one of its constituents?

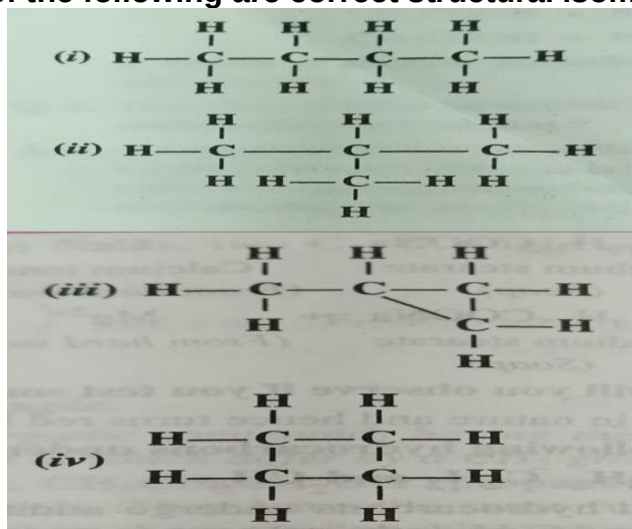
(a) Brass

(b) Bronze

(c) Amalgam

(d) Steel

5. Which of the following are correct structural isomers of butane?



(a) (i) and (ii)

(b) (ii) and (iv)

(c) (i) and (iii)

(d) (ii) and (iv)

6. During cellular oxidation of glucose, ATP is produced along with formation of other products in this reaction. Which of the following event is associated with production of maximum ATP molecules per molecule of Glucose during this process? Synthesis of

(a) Ethanol in yeast

(b) Lactic acid in muscle cells

(c) Carbon dioxide in yeast cells

(d) Carbon dioxide in human cells

7. If salivary amylase is lacking in the saliva, which of the following events in the mouth cavity will be affected?

(a) Proteins breaking down into amino acids

(b) Starch breaking down into sugar

(c) Fats breaking down into fatty acids and glycerol

(d) Absorption of vitamins

8. Choose the event that does not occur in photosynthesis.

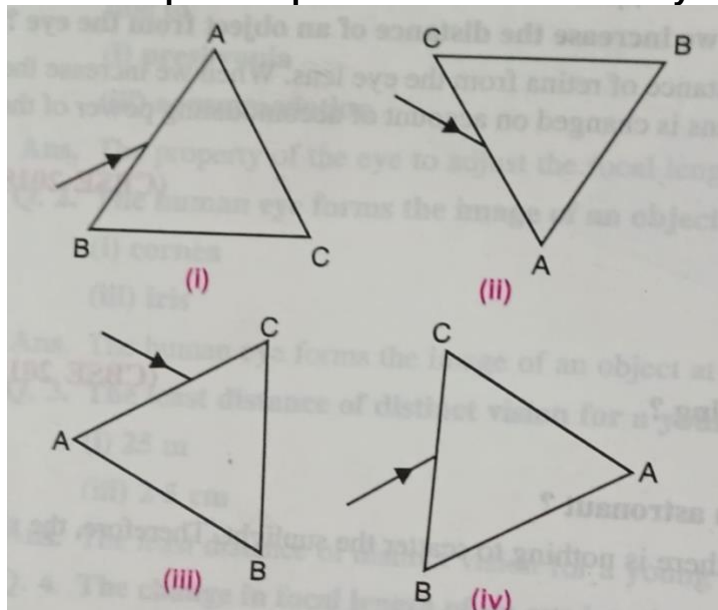
(a) Absorption of light energy by chlorophyll

(b) Reduction of carbon dioxide to carbohydrate

(c) Oxidation of carbon to carbon dioxide

(d) Conversion of light energy to chemical energy

9. When a person is suffering from severe cold, he or she can not
- Differentiate the taste of an apple from that of an ice cream
 - Differentiate the smell of a perfume from that of an agarbatti
 - Differentiate red light from green light
 - Differentiate a hot object from a cold object
10. The number of chromosomes in parents and offsprings of a particular species remains constant due to
- Doubling of chromosomes after zygote formation
 - Halving of chromosomes during gamete formation
 - Doubling of chromosomes after gamete formation
 - Halving of chromosomes after gamete formation
11. The plants that produce similar traits generation after generation are termed
- Pure line plants
 - Heterogenous plants
 - Reciprocal plants
 - Hybrid plants
12. Which part of the brain is the seat of intelligence and memory
- cerebrum
 - cerebellum
 - medulla oblongata
 - pons
13. A prism ABC (with BC as base) is placed in different orientation. A narrow beam of white light is incident on the prism as shown in figure. In which of the following cases, after dispersion, the third colour from the top corresponds to the colour of sky?



- (i)
 - (ii)
 - (iii)
 - (iv)
14. When three identical bulbs of 60 W – 200 V rating are connected in series to 200 V supply, the power drawn by them will be:
- 60 W
 - 180 W
 - 10 W
 - 20 W

15. A straight wire of diameter 0.5 mm carrying a current of 1 A is replaced by another wire of 1 mm diameter. The strength of the Magnetic field far away is
- (a) Twice the earlier value
 - (b) One half the earlier value
 - (c) One quarter of the earlier value
 - (d) Same as the earlier value
16. Disposable plastic plates should not be used because
- (a) They are made of materials with light weight
 - (b) They are made of toxic materials
 - (c) They are made of biodegradable materials
 - (d) They are made of non-biodegradable materials

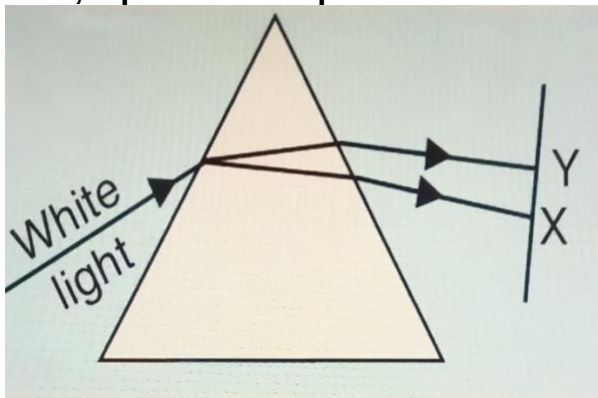
Questions numbers 17 to 20 consist of two statements- Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below

- (A). Both A and R are true, and R is the correct explanation of A.
 - (B). Both A and R are true, but R is not the correct explanation of A.
 - (C). A is true but R is false.
 - (D). A is false but R is true.
17. Assertion: When copper is heated in air, it turns black.
Reason: Copper reacts with H_2S gas of the air forming black CuS .
18. Assertion: IUCD are contraceptive devices made of copper, plastic, or stainless steel.
Reason: Copper-T prevents implantation of fertilized egg in uterus.
19. Assertion: The direction of magnetic field, force and current are Mutually perpendicular (Fleming's left hand rule)
Reason: Fleming's left-hand rule is used to measure current.
20. Assertion: UV radiation causes photodissociation of ozone into O_2 and O , thus causing damage to the stratospheric ozone layer.
Reason: -Ozone hole is resulting in global warming and climate change.

Section- B

Question No. 21 to 26 are very short answer questions

21. In the figure given below, a narrow beam of white light is shown to Pass through a triangular glass prism. After passing through the Prism, it produces a spectrum YX on the screen.



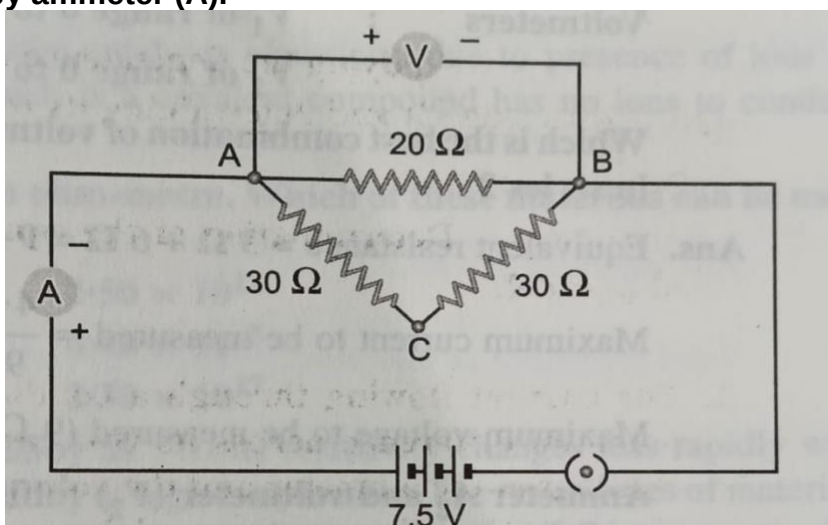
- (a) State the colour seen (i) at X, (ii) at Y.
 (b) Why do different colours of white light bend through different angles with respect to the incident beam of light?

OR

Identify the defect of vision in each of the following cases and suggest its corrective measures:

- (i) The eye lens has become milky and cloudy;
 (ii) Ciliary muscles have weakened.

22. In the circuit shown in the figure below. Find the current recorded by ammeter (A).



23. A concave lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object placed from the lens? Draw the ray diagram also.

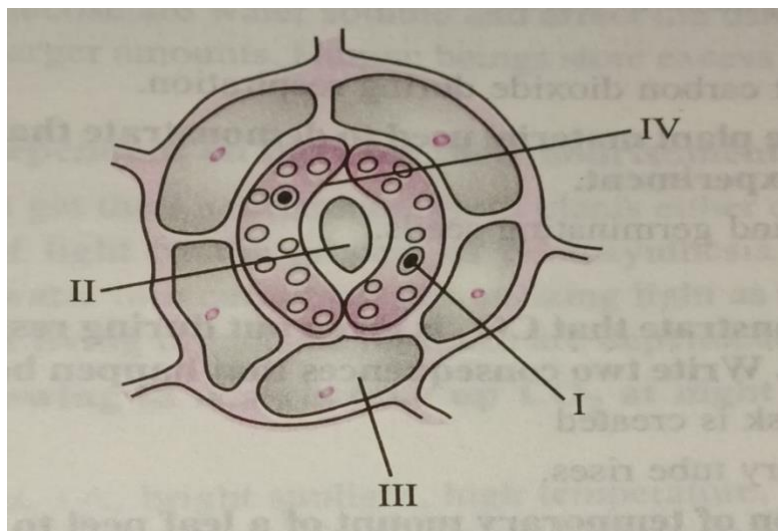
OR

An object is placed at a distance of 10 cm from a convex mirror of Focal length 15 cm. Find the position and nature of the image.

24. 2 g of ferrous sulphate crystals are heated in dry boiling tube. Write complete balanced chemical equation for the reaction and name the products formed.
 25. How would you distinguish between baking powder and washing soda by heating? Explain with the help of suitable chemical equations.
 26. Give reasons for the following.
 (i) Presence of cartilaginous rings in trachea.
 (ii) Why is small intestine in herbivore animals longer than in carnivore animals?

OR

A student after examining temporary mount of epidermal leaf Peel, has shown the following diagram and labelled four parts as I to IV. Name these parts.



Section- C

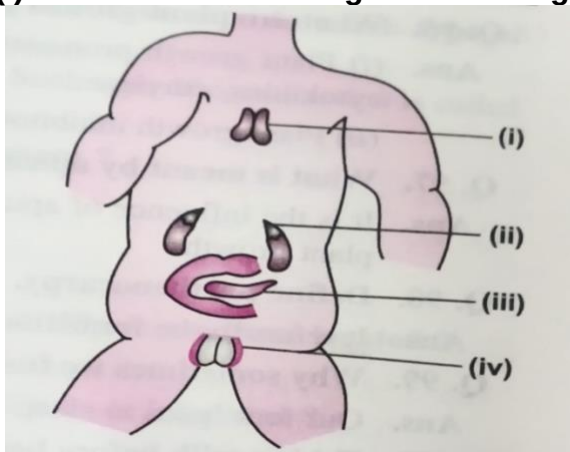
Questions No. 27 to 33 are short answer questions

27. (i) What are ozone depleting substances (ODS)?
 (ii) What are the harmful effects of ozone depletion on living Beings on Earth?
28. Students to attempt either A or B.
- A. A teacher asks his students to identify a metal, M. He gives them the following clues to help them.
- (P) Its oxide reacts with both HCL and NaOH.
 - (Q) It does not react with hot and cold water but reacts with Steam.
 - (R) It can be extracted by electrolysis of its ore.
- (a) Identify the metal.
 - (b) Write the chemical equations for the reaction of the metal with HCl and NaOH.
 - (c) What would happen if the metal is reacted with iron oxide?
- B. Explain the following
- (a) Reactivity of Al decreases if it is dipped in HNO_3 .
 - (b) Carbon cannot reduce the oxides of Na and Mg.
 - (c) NaCl is not a conductor of electricity in solid state whereas it does conduct electricity in aqueous solution as well as in molten state.
29. Alkanes are saturated compounds of carbon and hydrogen that can be represented by the general formula $\text{C}_n\text{H}_{2n+2}$ where 'n' is the numbers of carbon atoms. An example of such a compound is ethane C_2H_6 . Maya has a compound of carbon and hydrogen which formula is C_3H_4 .
- (i) What is true about the type of flame this compound will give on combustion?
 - (ii) Draw all the possible straight chain structures of this Compound.

OR

What happens when (write chemical equation to each case)

- (i) Ethanol is burnt in air
 - (ii) Ethanol is heated with excess conc. H_2SO_4 at 443 K.
 - (iii) A piece of sodium is dropped into ethanol.
30. (i) Label the endocrine glands in the given figure.



- (ii) Answer the following
 - (a) Deficiency of which hormone causes diabetes mellitus?
 - (b) Deficiency of which hormone causes dwarfism?
 - (c) Over secretion of which hormones causes exophthalmic goiter?
31. Answer the following questions
- (i) Define a gene.
 - (ii) Define the terms homogametic and heterogametic.
 - (iii) Distinguish between the acquired traits and the inherited traits.
32. A person needs a lens of power - 4.5 D for correction of his vision.
- (a) What kind of defect in vision is she suffering from?
 - (b) What is the focal length of the corrective lens?
 - (c) What is the nature of the corrective lens?
33. Why does a magnetic compass needle pointing North and south in the absence of a nearby magnet get deflected, when a bar magnet Or current carrying loop is brought near it. Describe some silent features of magnetic lines of field concept.

Section- D

Questions No. 34 to 36 long answer questions.

34. Carry out the following conversions, stating the condition(s) for each.
- (i) Ethanol $\rightarrow$ Ethene
 - (ii) Ethene $\rightarrow$ Ethane
 - (iii) Ethane $\rightarrow$ Chloroethane
 - (iv) Ethanol $\rightarrow$ Ethanoic acid
 - (v) Ethanoic acid $\rightarrow$ Ethyl ethanoate

OR

- (i) Distinguish between a saturated and an unsaturated hydrocarbon by flame test.
- (ii) Name the products formed when ethane burns in air. Write the

balanced chemical equation for the reaction showing the types of energies liberated.

(iii) Why is reaction between methane and chlorine in the presence of sunlight considered substitution reaction?

35. Attempt either option A or B,

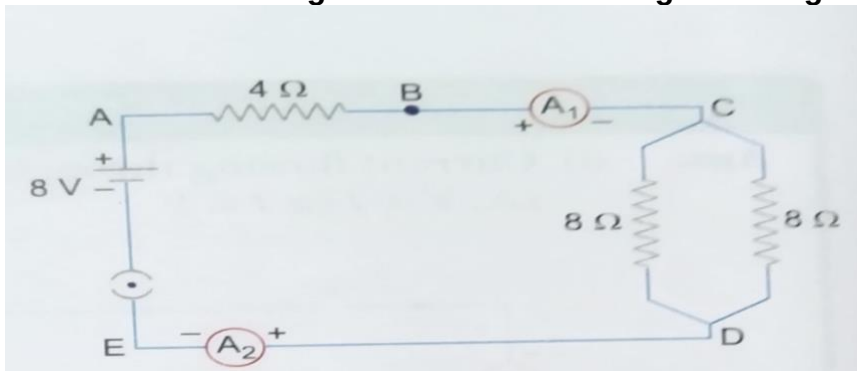
A. Describe the structure and functioning of nephron with the help of a well labelled diagram.

B. Draw a well labelled diagram of human heart, and write the functions of the following

(i) Pulmonary artery (ii) pulmonary vein (iii) Aorta

(iv) Septum (v) Left ventricle

36. Find out the following in the electric circuit given in figure



(i) Effective resistance of two 8 Ohm resistors in the combination.

(ii) Current following through 4 Ohm resistor.

(iii) Potential difference across 4 Ohm resistor.

(iv) Power dissipated in 4 Ohm resistor.

(v) Difference in ammeter reading if any.

Section- E

37. Read the passage carefully and answer the question bellow.

Most of the common compounds which are used in our everyday life is acidic, basic or salt. It is dangerous to taste them. Their Nature can be tested with suitable indicators. In fact salts are formed by the neutralization of acids and bases or by certain other reactions. Different acids react with metals to different extents while some do not react at all.

(i) Bitter gourd is

(a) Acidic

(b) Basic

(c) Neutral

(d) Anyone of these

(ii) Which of the following is a weak acid?

(a) Oxalic acid

(b) Hydrochloric acid

(c) Nitric acid

(d) Sulphuric acid

(iii) Which of the following salt solution in water is basic in nature?

(a) KNO_3

(b) NaCl

(c) Na_2CO_3

(d) CuSO_4

(iv) Milk of magnesia is the name given to the compound

38. Human male reproductive system comprises a pair of testes, scrotum, vas deferens, urethra, penis and a few glands. Testes are

the primary reproductive organs and are termed male gonads. These produce sperms or male gametes. The latter are temporarily stored in the seminal vesicle and where mucus and watery alkaline fluid containing the sugar is mixes with the sperms.

- (i) Where are the two testes, in the male reproductive system, located?
- (ii) Which sugar is present in the alkaline watery fluid that mixes with the sperms along with mucus in the seminal vesicle?
- (iii) In human males, testes produce sperm from -----
- (iv) Name the hormone produced by testes.

39. A student took three concave mirrors of different focal length performed the experiment to see the image formation by placing an object at different distance s with this mirrors shown in the following table

Case No.	Object distance	Focal length
I	45 cm	20 cm
II	30 cm	15 cm
III	20 cm	30 cm

- (a) List two properties of the image formed in case I.
(b) In which one of the case given in the table the, the mirror will form real image of size and why?
(iii) Name the type of mirror used by dentist. Give reason why do they The use such type of mirror.

OR

The image of an object formed by a convex lens may be real /virtual, erect/inverted, smaller/larger than the object. It would depend upon distance of the object from the lens. However, the image of an object formed by a concave lens is always virtual, erect and smaller in size than the object.

- (a) To obtain image of an object on a screen, we require a
- | | |
|---|---|
| (i) Convex lens. | (ii) Concave lens |
| (iii) Concave lens of large focal length. | (iv) Concave lens of small focal length |

- (b) Where should an object be placed in front of a concave lens to obtain a virtual image of the object?**

- | | |
|-------------------------------|---------------------------------|
| (i) Infinity. | (ii) Principal focus |
| (iii) Very close to the lens. | (iv) Any distance from the lens |

- (c) Where should an object be placed in front of a convex lens to obtain image of the size of the object?**

- (i) At focus F. (ii) At 2F
(iii) Beyond 2 F. (iv) None of these

- (d) To obtain an image smaller than the size of the object, we may used a**

- (i) Convex lens only. (ii) Concave lens only
(iii) Either a convex lens or a concave lens (iv) Cannot say
